Relational Architecture of Dimensionality via Reduction of R

Conduction, Optical Guiding, and Magnetism as Constraints on Degrees of Freedom

Luis Diego Mata Sánchez

Abstract

We propose an ontological–operational framework in which observable physical properties (electrical conduction, optical guiding, and magnetism) are interpreted as architectures of constraint on degrees of freedom, acting as *bridges* that reduce the relational radius R of information within a system. In this view, *distance* between subsystems is not a pre-existing geometric container, but an operational measure of mismatch relative to a universal reference (c) and an experimental scale (ω_*). Functional dimensions emerge when relational information cannot be accommodated within the previous framework and requires additional independent channels. We provide minimal definitions, falsifiable hypotheses, and a mapping dictionary for materials: conductor, optical fiber, and ferromagnet.

1 Motivation: dimension as space for new information

We adopt the operational definition: a *dimension* is not a pre-given container, but a functional space required to store information that did not fit within the previous descriptive frame. Accordingly, a system's effective dimensionality is associated with the number of independent channels required to describe its internal relations.

2 Central variable: relational radius R as a measure of informational difference

Consider two subsystems i and j with comparable relational states. Let $\Delta\phi_{ij}$ denote a measurable mismatch (in the appropriate variable: optical phase, effective electronic phase, etc.). We define the *relational distance* operationally as

$$R_{ij} \;=\; \frac{c}{\omega_*}\,|\Delta\phi_{ij}|, \tag{1}$$

where c plays the role of a universal reference (not a temporal ontology) and ω_* fixes the experimental class (e.g. transport versus optical response).

Illustrative numerical example (optional)

To fix scale: if one takes $\omega_* = 2\pi \times 10^{15}\,\mathrm{s}^{-1}$ (optical order) and $|\Delta\phi| = 10^{-2}\,\mathrm{rad}$, then

$$R \sim \frac{3\times 10^{8}}{2\pi\times 10^{15}}\,10^{-2} \approx 5\times 10^{-10}\,\mathrm{m},$$

i.e. a sub-nanometer order. This estimate is illustrative: ω_* must be pre-registered per experimental class and not tuned per sample.

Reading. R is not a coordinate; it is a measure of informational discrepancy between states/paths. Reducing R corresponds to reducing effective mismatch between subsystems.

3 Material as an operator that constrains degrees of freedom

Let $\mathcal{M}$ denote the *material architecture* (conductor, waveguide, ferromagnet, etc.). We postulate that its core physical role is to restrict the accessible state space and thus modify mismatch evolution:

$$\mathcal{M}: \ \Delta\phi_{ij}(t) \mapsto \Delta\phi'_{ij}(t) \quad \Rightarrow \quad R_{ij} \mapsto R'_{ij}. \tag{2}$$

We propose two minimal representations:

3.1 Static model (effective contraction)

$$R' \ = \ \beta(\mathcal{M})\, R, \qquad 0 < \beta(\mathcal{M}) < 1, \tag{3}$$

where β summarizes the net "sieve" action of the material on allowed degrees of freedom.

3.2 Dynamic model (mismatch relaxation)

$$\frac{d}{dt}R \ = \ -\Gamma(\mathcal{M})\, R + \xi(t), \tag{4}$$

where $\Gamma(\mathcal{M})$ quantifies the material's capability to suppress dephasing channels and $\xi(t)$ represents residual environmental fluctuations.

Minimal consistency note. In a near-equilibrium linear regime, the dynamic model induces an effective contraction over finite observation windows; hence it is reasonable to explore order-of-magnitude relations of the form $\Gamma^{-1} \sim \tau(\beta)$ (not necessarily universal), where τ is a relaxation timescale associated with the material architecture.

4 Effective dimensionality as the count of active degrees of freedom

We define functional effective dimensionality as

$$D_{\text{eff}} \ = \ N_{\text{dof}}^{(\text{active})}, \tag{5}$$

i.e. the number of dynamical channels required to describe relational organization. In this view, a material is "efficient" for conduction/guiding not because it "carries energy", but because it *deactivates* channels that amplify $\Delta\phi$ and therefore R.

5 Three guiding cases: conductor, optical fiber, and ferromagnet

5.1 Conductor (wire)

Interpretation: an electrostatic potential gradient is not taken as a fundamental energy carrier, but as a *constraint condition* that selects/alines electronic states and reduces scattering channels. The conductor acts as $\mathcal{M}$ that channels compatible states and minimizes R relative to the environment. Operationally, it functions as an "ontological sieve": it retains, reflects, or transmits modes according to structural compatibility.

5.2 Optical fiber

Interpretation: waveguiding is a geometric architecture that restricts transverse modes, suppresses phase leakage, and preserves propagation coherence. In terms of (1), the fiber reduces the growth of $|\Delta\phi|$ with geometric distance, producing a "smaller relational radius" than free-space propagation.

5.3 Ferromagnet (iron)

Interpretation: magnetism introduces a collective rotational mode that reduces incoherent degrees of freedom and generates alignment (order) that stabilizes relational transfer. Net effects include suppression of contributions to $\xi(t)$ in (4) and/or increased $\Gamma(\mathcal{M})$.

6 Minimal falsifiable hypotheses

We propose three independent tests:

1. **Material contraction:** measure $\beta(\mathcal{M})$ across architectures (material, geometry, impurities) and test whether R-reduction predicts conduction/guiding trends without re-tuning ω_* per sample.

2. **Layer separation:** if the mechanism is dof-constraint, geometric modifications that open extra channels must increase D_{eff} and degrade performance via increased R. This connects to operational channel-control programs (e.g. a parameter κ in 2D models), but does not depend on them.

3. **Coherence threshold:** there exists an operational regime where the system ceases to be a "bridge": when $\Gamma(\mathcal{M})$ falls below a threshold (or ξ dominates), R no longer contracts and transport becomes diffusive/dispersive.

7 Optional connection to harmonic partition

This manuscript does not require energy/time as ontological primitives. However, to connect with a harmonic-partition formalism, one may interpret the system as having "projected" (propagating/coherent) modes and "confined" (identity/retentive) modes satisfying an orthogonal scale relation. This layer is declared optional; its integration is left as a future verifiable mapping.

8 Conclusion

We reinterpret conduction, optical guiding, and magnetism as mechanisms of constraint on degrees of freedom that reduce the relational radius R of information. Rather than postulating dimensions, energy, or time as primary containers, effective dimensionality emerges as a count of channels required to store new relational information. The approach is operational: it defines R via mismatch and demands falsifiable tests through the response of $\beta(\mathcal{M})$ or $\Gamma(\mathcal{M})$ under controlled changes of material and geometry.

References

[1] L. D. Mata Sánchez, *Emergent Dimensionality via Electromagnetic Interaction: Gradient, Rotation, and Propagation as Relational Modes*, manuscript.

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